YUKUN GUO

PH.D. CANDIDATE

Portland, Oregon | 503.380.0669 | guoyukn@gmail.com | LinkedIn | GitHub | Google Scholar

POSITIONING STATEMENT

Seeking a Research Scientist/Medical Image Analysis role where my deep expertise in biomedical imaging, AI algorithm development and clinical tool deployment can drive advanced diagnostics and clinical value.

PROFESSIONAL PROFILE

Ph.D. Candidate in Biomedical Engineering with 8+ years of dedicated research and software development experience in optical coherence tomography (OCT), OCT angiography (OCTA) and medical AI pipelines. Recognized for designing and deploying deep learning systems for segmentation, classification, reconstruction, denoising, vessel and tissue boundary analysis, and quantitative biomarker extraction. Inventor on three U.S. patents and author of 40+ peer reviewed publications. Proficient in translating novel imaging algorithms into robust production ready tools licensed to both industry and clinical research partners. Adept at partnering with clinicians, imaging specialists, regulatory teams and software engineers to accelerate clinical trial readiness of imaging biomarkers. Seeking to bring this rare blend of algorithmic rigor, imaging domain depth and translational deployment experience to a research scientist role in medical image analysis.

CORE COMPETENCIES

- Cross-functional collaboration: clinicians, post-docs, industry
- Scientific writing, peer-review and regulatory-aware design
- GPU acceleration (CUDA), ONNX, TensorFlow/Keras/PyTorch
- GUI/UX applications (Qt, PySide6) for clinical research
- Multi-device compatibility & imaging-vendor support
- OCT / OCTA image processing & retinal biomarkers
- Multimodal imaging & vascular tissue quantification
- Large-scale image data pipelines & batch processing
- Clinical image-software deployment & licensing
- C++/Python/MATLAB software development
- Deep Learning & CNN/U-Net architectures
- 3D image segmentation & reconstruction

EDUCATION

PH.D. CANDIDATE, BIOMEDICAL ENGINEERING, 2022–PRESENT; *Oregon Health & Science University, Portland, OR*
M.S., COMPUTER SCIENCE, 2014–2017; *University of Jinan*
B.S., COMPUTER SCIENCE, 2009–2013; *University of Jinan*

WORK EXPERIENCE

OREGON HEALTH & SCIENCE UNIVERSITY, PORTLAND, OR | 2022–PRESENT

GRADUATE RESEARCHER

- Designed a 3D OCTA signal restoration model; developed automated grading of age-related macular degeneration (AMD) severity.
- Developed advanced deep-learning segmentation models for geographic atrophy and multi-decoder CNNs for simultaneous segmentation of drusen, retinal pigment epithelium and vascular structures — enabling AI-aided segmentation of four drusen types in volumetric OCT.
- Built a deep-learning model to segment nonperfusion areas (NPAs) across three retinal vascular plexuses on OCTA volumes.
- Co-invented U.S. patent on automated detection of avascular and signal-reduction regions in retinal OCTA (US11974811, 2024).
- Collaborated with clinical and imaging teams to translate AI algorithms into tools for clinical trials and early disease detection.
- Developed and deployed a comprehensive OCT/OCTA analysis software platform (desktop and research-grade), embedded with AI models for clinical applications — including a commercialized retinal-layer segmentation module licensed to Genentech Inc.
- Engineered annotation and visualization tools (web portal, research-team website) supporting group researchers' workflows.

Key Achievements:

- Licensed software platform to Genentech Inc. and deployed at 30+ researchers across 40+ projects — significantly improving reproducibility and throughput of retinal-image analysis.
- Co-inventor on patent US11974811 B2.

OREGON HEALTH & SCIENCE UNIVERSITY, PORTLAND, OR | 2019–2022
RESEARCH ASSISTANT

- Developed multiple end-to-end CNNs for OCTA image segmentation; e.g., "Automated Segmentation of Retinal Fluid Volumes from Structural and Angiographic OCT Using Deep Learning (TVST, 2020)".
- Created the MEDnet AI system for automated detection of avascular areas in OCT angiography (Biomed. Opt. Express, 2018/2019), later adopted in clinical research for diabetic retinopathy monitoring.
- Designed deep-learning frameworks for wide-field OCTA nonperfusion quantification (e.g., "Quantification of Nonperfusion Area in Montaged Wide-field OCTA Using Deep Learning", Ophthalmology Science, 2021).
- Contributed to open-source retinal-image reconstruction & segmentation tools ("An Open-Source Deep Learning Network for Reconstruction of High-Resolution OCT Angiograms", TVST 2021).
- Developed and deployed retinal layer segmentation software used clinically at Casey Reading Center (Portland, OR).

Key Achievements:

- Published 20+ peer-reviewed papers (2019–2022) advancing reproducibility in AI-OCT research.
- Technology used in a clinical-reading center environment.

OREGON HEALTH & SCIENCE UNIVERSITY, PORTLAND, OR | 2017–2019
VISITING RESEARCHER

- Initiated collaborative research on automated OCTA analysis; produced early publications on "Automated Segmentation of Retinal Layer Boundaries and Capillary Plexuses" (Biomed. Opt. Express, 2018).
- Developed prototype deep-learning algorithms for retinal vasculature segmentation which evolved into patented clinical-tools.

Key Achievements:

- Established early foundational algorithms leading to later commercial and clinical translational outcomes.

PATENTS AND LICENSED INTELLECTUAL PROPERTY

- US 11974811 B2 – Detecting Avascular and Signal Reduction Areas in Retinas Using Neural Networks. Licensed to Optovue/Visionix Inc. and IFOCUS Imaging LLC.
- US 11717155 B2 – Identifying Retinal Layer Boundaries from Reflectance OCT Images.
- US 11263747 B2 – Detecting Avascular Areas Using Neural Networks on OCTA Images.
- Licensed Software – COOL Lab Optical Coherence Tomographic Angiography Reading Toolkit (COOL-ART) licensed to Genentech Inc.
- Licensed Technique – Automated Segmentation of Retinal Fluid from OCT/OCTA Using Deep Learning ("ReF-Net") licensed to IFOCUS Imaging LLC.

AWARDS AND HONORS

- Outstanding Reviewers, Optical Society of America (OSA), 2021
- Travel Award, International Ocular Circulation Society (IOCS), Kyoto, Japan, 2021
- Retina Research Foundation / Joseph M. & Eula C. Lawrence Travel Grant, Association for Research in Vision and Ophthalmology (ARVO), Vancouver, Canada, 2019

ACADEMIC ACTIVITIES

- Peer-Reviewer for top journals including IEEE Transactions on Medical Imaging (TMI), npj Digital Medicine, Biomedical Optics Express (BOE), Translational Vision Science & Technology (TVST), Ophthalmology Science.
- Conference Presentations (oral & poster) at ARVO Imaging in the Eye Conference (2018-2024) covering automated volumetric segmentation, geographic atrophy, wide-field OCTA etc.
- Lead Projects: OCT/OCTA Analysis Software (2019–Present) — production-ready platform, supported 30+ researchers; AI-derived layer-specific OCT biomarkers for classification of geographic atrophy; Deep-learning based AMD severity grading using multimodal data; AI-aided segmentation of four types of drusen in volumetric OCT; Multi-plexus non-perfusion area segmentation in wide-field OCTA using CNN.

PROJECT EXPERIENCE

OCT/OCTA Analysis Software Platform
Developer & Architect | Jan 2019 – Present

- Designed and engineered a comprehensive, production-ready software suite for advanced OCT/OCTA image analysis, licensed to Genentech Inc. and deployed for 30+ researchers. This platform integrates 8+ years of algorithm development across retinal segmentation, fluid detection, atrophy mapping, and device-agnostic data handling.
- Supported OCT/OCTA data ingestion from all major commercial devices (Optovue, Zeiss, Topcon, Heidelberg, etc.).
- Embedded deep-learning models for segmentation of 11 retinal layers, retinal fluid, drusen, and geographic atrophy.
- Delivered visualization tools, batch processing pipelines, and advanced annotation features for image-labeling and area/volume quantification.
- Enabled high-throughput, reproducible, multi-center research across 40+ projects.

Impact: Drove the clinical translation of research algorithms into real-world tools used in R&D and trials by industry and academic collaborators.
Skills: Deep Learning, PyTorch, Qt, C++, Python, MATLAB, Image Segmentation, GUI/UX, Multithreading

AI-Derived OCT Biomarkers for Geographic Atrophy Classification
Lead Developer | 2022 – Present

- Created a novel deep-learning framework for voxel-level quantification of layer-specific atrophic features in dry AMD using OCT volumes.
- Automated quantification of EZ–RPE disruptions, hyper-transmission defects, and inner nuclear layer thinning.
- Achieved voxel-level F1-scores of 0.76 (cRORA), 0.64 (EZ–RPE loss), and 0.69 (hyperTDs).

Impact: Provided new biomarkers to support objective geographic atrophy progression metrics in clinical trials.
Skills: PyTorch, 3D CNNs, Biomarker Analysis, OCT

Multimodal Deep Learning for AMD Severity Grading
Research Lead | Aug 2025 – Present

- Designed a hierarchical classification system integrating 3D OCT, OCTA via cross-modal feature fusion.
- Developed a multimodal AI architecture for improved accuracy in AMD severity staging.

Impact: Enabled early-stage disease detection and enhanced clinical decision-making through multi-imaging analysis.
Skills: PyTorch, 3D CNNs, Multimodal AI, Medical Image Fusion

AI-Aided Segmentation of Four Drusen Types in Volumetric OCT
Primary Developer | Nov 2024 – May 2025

- Built a CNN-based system combining segmentation and classification to detect and categorize soft drusen, subretinal drusenoid deposits (SDD), drusenoid pigment epithelial detachment (DPED), and cuticular drusen.
- Achieved Dice score of 0.74 and critical success index of 0.69.

Impact: Improved quantification of drusen phenotypes for AMD progression tracking and treatment evaluation.
Skills: TensorFlow, 3D Segmentation, OCT Biomarker Development

Widefield OCTA Nonperfusion Area Segmentation Using CNNs
Model Architect | Jan 2024 – Jul 2024

- Developed a parallel U-Net–based CNN for multi-plexus segmentation of nonperfusion areas across superficial (SVC), intermediate (ICP), and deep (DCP) capillary plexuses.
- Achieved F-scores between 0.84–0.87 across plexuses.

Impact: Enabled high-resolution ischemia mapping for diabetic retinopathy staging and treatment assessment.
Skills: TensorFlow, Widefield Imaging, OCTA, Retinal Plexus Segmentation

ADDITIONAL INFORMATION

Languages: English, Chinese

Technical Proficiencies: Windows OS, Linux, Microsoft Office (Word, Outlook, Excel, PowerPoint), ChatGPT, Copilot, Git/GitHub

Interests: photography, hiking, coding, emerging technologies